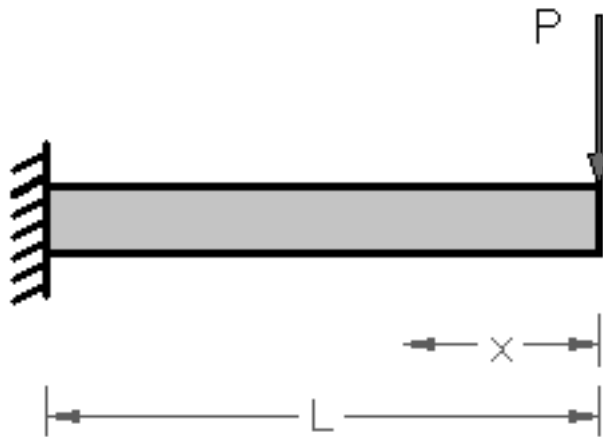


Vibration Analysis of a Cantilever Beam



MECH 460
Finite Element Analysis

Prof. Behrooz Yousefzadeh

We certify that this submission our original work and meets the faculty's expectation
of originality.

Mohammad Umer (40150744)

Ha Tuan Khanh Nguyen (40189976)

Concordia University

Montreal, QC

Contents

Abstract	4
Introduction	4
Methodology [1][2]	5
Question 1	5
Question 2	5
Question 3	6
Question 4	6
Result.....	6
Question 1	6
Question 2	10
Question 3	11
Question 4	13
Conclusion	16
References.....	17
Appendix A	18
Running Python Code	18

Figures

Figure 1: First 4 Natural Frequencies and Relative Error for 4, 8 , 16, 32, 64 and 128 elements ...	7
Figure 2: Code of Beam Element M and K matrix.....	8
Figure 3: Max Relative Error vs No. of Elements.....	8
Figure 4: Steady-State Response at Motor Position due to Harmonic Force	10
Figure 5: Dynamic Bending Stress Code Snippet	11
Figure 6: Bending Stress at Root	11
Figure 7: Steady-State Response at Tip due to Harmonic Force	12
Figure 8: Mass added to Cantilever Beam Code	13
Figure 9: Steady State Response at Tip due to Harmonic Force with added Mass	13
Figure 10: Compression Spring Characteristics.....	14
Figure 11: Addition of Spring to Cantilever Beam Code	14
Figure 12: Maximum Displacement at Tip with Spring Added	15
Figure 13: Steady State Response at Tip due to Harmonic Force with added Spring.....	15

Abstract

The problem involves a prismatic cantilever beam being fixed at one end, given its total length, density, and cross-section dimensions. This information is provided to give the solution using Finite Element Analysis (FEM) on a dynamic structure, which is implemented by computer algorithms. Four main questions will be addressed in this project. The first one asks for the first four natural frequencies of the cantilever and the number of meshing elements required to have a relative error of 0.1%. Secondly, a DC motor with force of 1kN amplitude is placed at a certain distance from the supported end of the motor. The problem is then to plot the frequency response of the vibration amplitude at one point, chosen to be where the motor is placed. The third problem asks for the maximum bending stress at the root of the beam at 32Hz frequency, and the maximum vertical displacement of the tip of the beam from 1Hz to 600Hz.

Introduction

In this project, the vibration response of a cantilever beam subjected to harmonic external force is investigated. The material model is typical of aluminum with $E = 68 \text{ GPa}$, $\rho = 2700 \frac{\text{kg}}{\text{m}^3}$, $A = 6 \times 10^{-4} \text{ m}^2$, $I = 5 \times 10^{-9} \text{ m}^4$, $L = 0.5 \text{ m}$. The primary object is then to:

1. Compute first 4 natural frequencies of this beam and find the number of meshing elements at which relative error is 0.1%.
2. Introduce harmonic force and plot frequency response from 1Hz to 600Hz
3. Calculate the maximum stress due to bending at the root of the beam at 32Hz, and find the maximum vertical displacement in the given range of frequency
4. Reduce maximum tip displacement to $\leq 8\text{cm}$ using mechanical elements like point mass, spring or dampers.

Methodology [1][2]

Question 1

1. Discretize the beam into elements.
2. Define shape function.
3. Identify element mass and stiffness matrix.

The element stiffness matrix $[K_e]$ and mass matrix $[M_e]$ are derived using the beam's flexural rigidity (EI) and mass per unit length (ρA) , respectively.

4. Assemble the global mass and stiffness matrix.
5. Apply boundary conditions.
6. Solve for $[K] - (\omega_n)^2[M]\{\phi_n\} = 0$

Question 2

Equation (2) is the governing equation of harmonic vibration analysis.

$$[K] - \omega^2[M] + [C]\{q(\omega)\} = \{F(\omega)\},$$

Where $[K]$: global stiffness matrix

$[M]$: global mass matrix

$\{q(\omega)\}$: frequency domain displacement vector

ω : frequency of harmonic force input, between 0 and 600Hz

$\{F(\omega)\}$: Force vector of amplitude 1000kN

$[C]$: diagonal damping ratio.

Then, the displacement vector is shown in equation 3.

$$\{q(\omega)\} = ([K] - \omega^2[M] + [C])^{-1} \cdot \{F(\omega)\}$$

The displacement vector has both real and imaginary parts and the overall amplitude is used for the following harmonic analysis.

In solving this problem, the global mass matrix and global stiffness matrix were already assembled for the previous part. So, only the new damping matrix needs to be introduced. After setting up, the problem can be solved according to equation number 3.

Question 3

The maximum moment in a beam according to Euler-Bernoulli theory is given by

$$M = EI \frac{d^2 v}{dx^2} = EI \frac{d\theta}{dx} = EI \frac{\theta_{node_2} - \theta_{node_1}}{\frac{Length\ of\ Beam}{No.\ of\ Elements}}$$

The corresponding stress in a beam element is then obtained using

$$\sigma = \frac{Mc}{I}$$

The maximum lateral displacement at the tip within the frequency range 0-600 Hz is obtained from the analysis in question 2.

Question 4

Simple mechanical elements like dashpots and springs were sourced from McMaster-Carr.

Dashpots are rated by energy capacity and no relevant documentation is available with respect to damping coefficients.

Hence analysis was done by introducing either a spring or mass element into the system.

Result

Question 1

The number of nodes is related to the number of elements by $n+1$ and these terms might be used interchangeably depending on scope of discussion.

Analytically the first 4 natural frequencies of a cantilevered beams are given using the following equation:

$$f_n = \frac{c_n^2}{2\pi L^2} \sqrt{\frac{EI}{\rho A}} \text{ where}$$

$$c_1 = 1.875, c_2 = 4.694, c_3 = 7.855 \text{ and } c_4 = 10.996$$

The first 4 natural frequencies obtained analytically are:

1 st	32.42 Hz
2 nd	203.1 Hz
3 rd	569.5 Hz
4 th	1115 Hz

Table 1: Natural Frequencies Exact Solution

```

Frequencies for 128-element mesh: [ 32.42745227 203.21937381 569.02013574 1115.05239458]
4 nodes: [ 32.43073834 203.88680455 576.11030751 1297.37883571]
Relative Error: [1.01335826e-02 3.28428697e-01 1.24603179e+00 1.63513788e+01]

8 nodes: [ 32.42756712 203.246844 569.59946788 1119.16899416]
Relative Error: [3.54163816e-04 1.35175058e-02 1.01812238e-01 3.69184408e-01]

16 nodes: [ 32.42745759 203.22071586 569.04935456 1115.26983479]
Relative Error: [1.64085341e-05 6.60396022e-04 5.13493565e-03 1.95004481e-02]

32 nodes: [ 32.42745244 203.21944752 569.02175255 1115.06452877]
Relative Error: [5.18978132e-07 3.62716700e-05 2.84138972e-04 1.08821734e-03]

64 nodes: [ 32.42745222 203.21937755 569.02022557 1115.0530687 ]
Relative Error: [1.51254026e-07 1.84116576e-06 1.57854682e-05 6.04565259e-05]

```

Figure 1: First 4 Natural Frequencies and Relative Error for 4, 8, 16, 32, 64 and 128 elements

The figure above shows the first 4 natural frequencies for 4, 8, 16, 32, 64 and 128 element meshes and the corresponding relative errors. General trend observed is that as the mesh is refined i.e. has more elements; the closer the FEA solution gets to the analytical solution.

The beam elements stiffness and consistent mass matrix are given below:

```

# --- Beam Element Matrices ---
def beam_element_stiffness(E, I, L):
    return np.array([
        [12*E*I/L**3, 6*E*I/L**2, -12*E*I/L**3, 6*E*I/L**2],
        [6*E*I/L**2, 4*E*I/L, -6*E*I/L**2, 2*E*I/L],
        [-12*E*I/L**3, -6*E*I/L**2, 12*E*I/L**3, -6*E*I/L**2],
        [6*E*I/L**2, 2*E*I/L, -6*E*I/L**2, 4*E*I/L]
    ])

def beam_element_mass(rho, A, L):
    coeff = rho * A * L / 420
    return coeff * np.array([
        [156, 22*L, 54, -13*L],
        [22*L, 4*L**2, 13*L, -3*L**2],
        [54, 13*L, 156, -22*L],
        [-13*L, -3*L**2, -22*L, 4*L**2]
    ])

```

Figure 2: Code of Beam Element M and K matrix

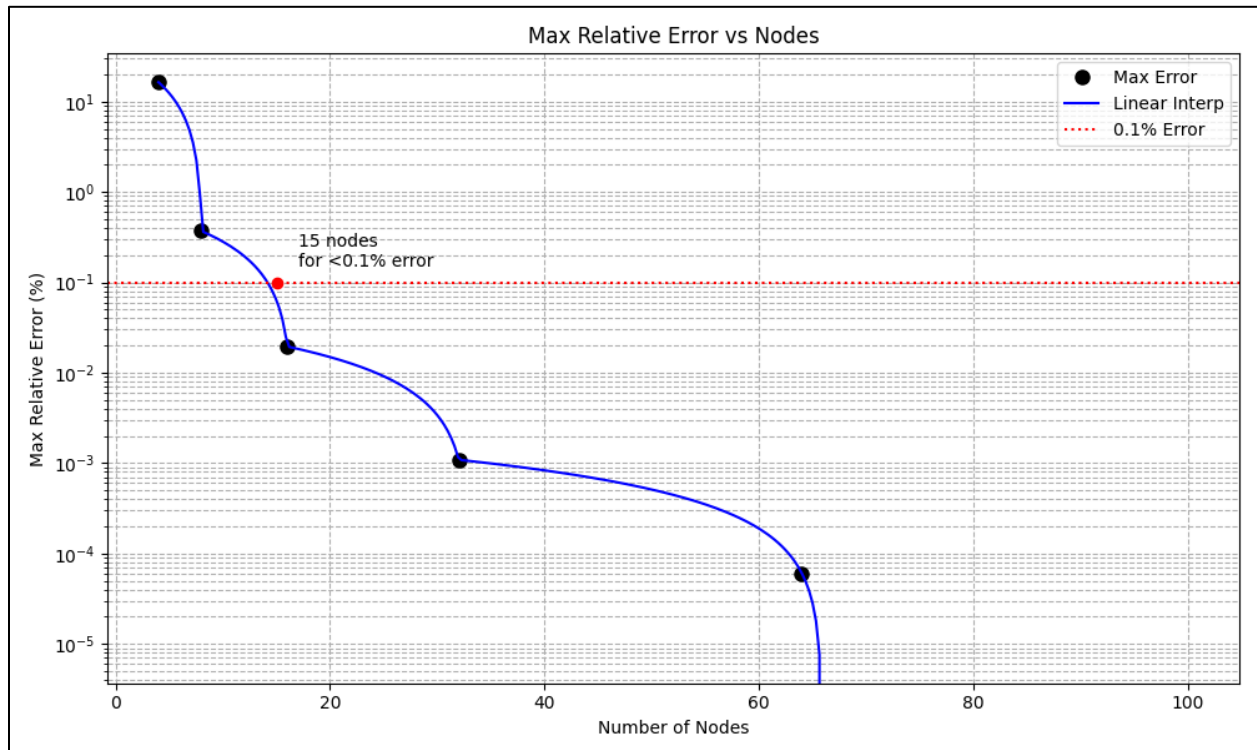


Figure 3: Max Relative Error vs No. of Elements

The above graph plots the maximum error within the first 4 natural frequencies when the number of elements is varied between 4, 8, 16, 32 and 64 nodes. For frequencies between these values a linear interpolation is performed. It is observed that around 15 / 16 elements the maximum error within the first 4 natural frequencies is within 0.1% of analytical values.

From figure 1, it is observed that even with 8 elements the maximum error observed drops to only 0.37%.

Also a general decrease in the error is observed as the number of elements / nodes is increased.

Question 2

For the following questions the beam is modeled with 128 elements and a linear viscous damping mechanism with a diagonal matrix constant = 0.008.

Since the length of the beam is 0.5m and it is divided into 128 elements and the motor is placed at 0.15m from fixed end which is not an integral multiple of $\frac{0.5}{128}$ m; a simplification made to aid with the analysis is that the point of application of the force is moved to the node closest to the actual point of application. This simplification will also improve if more refined mesh is used. Currently the force falls on the 38.4 (38th) element which is 0.148m from the fixed end.

Plotting beam shapes for all 600 simulated frequency intervals would take up unnecessary space; the displacement of the beam at the motor location is plotted instead in the figure below.

Displacement peaks coincide with first 3 natural frequencies of the cantilever beam. The peaks are 0.55 cm, 0.26 cm and 0.12 cm at first, second and third natural frequency respectively.

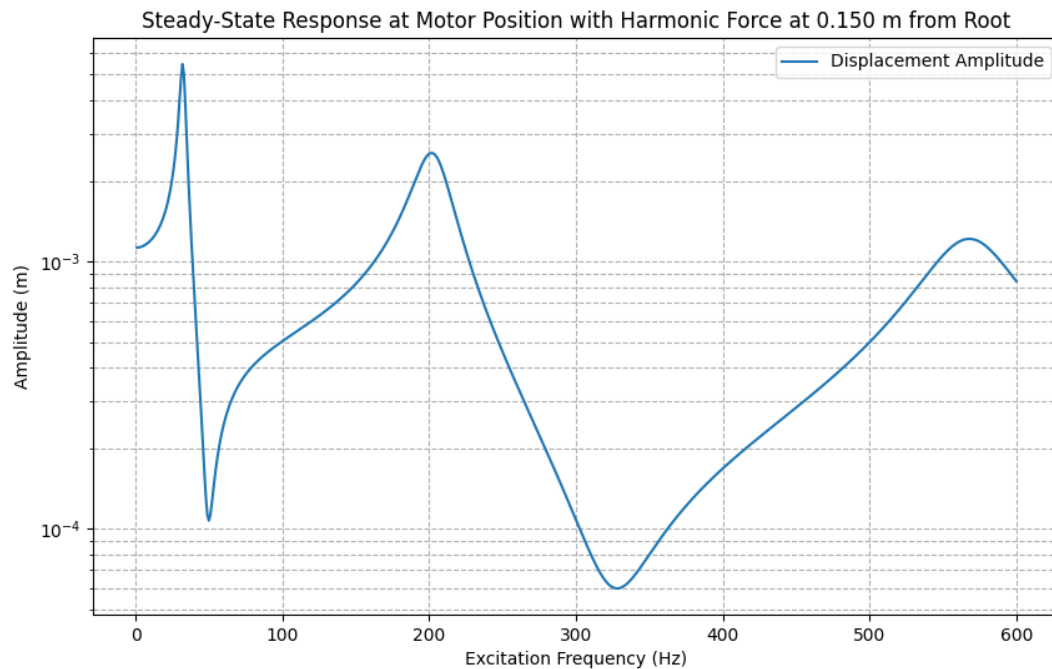


Figure 4: Steady-State Response at Motor Position due to Harmonic Force

Using the W variable inside the frequency response function, the steady state beam shapes can be drawn out for all frequencies within testing range. The W variable is a numpy array containing absolute value of the imaginary displacements returned by solving the forced vibration eigenvalue problem.

Question 3

```
def compute_dynamic_bending_stress(E, I, L_total, response, node_idx):  
    dx = L_total / 127  
    y_max = 0.005 # Outer fiber distance for rectangular section approx.  
    stress = E * (response[1]/dx) * y_max  
    print(f"\n Stress at root: {stress} Pa")  
    return stress
```

Figure 5: Dynamic Bending Stress Code Snippet

The maximum bending stress at the root element is obtained as outlined in the methodology section; the value is given in the figure below and is approximately 650 MPa.

Dynamic bending stress at root: 6.49841e+08 Pa

Figure 6: Bending Stress at Root

Using the calculator / formula on The Engineering Toolbox, the maximum static bending stress at the root for a 1000N force is 150 MPa; this value is a minimum for the bending stress at the root and does not take into account dynamic amplification at resonance since analysis is carried out at 32 Hz which is in the vicinity of the beams first natural frequency. [3]

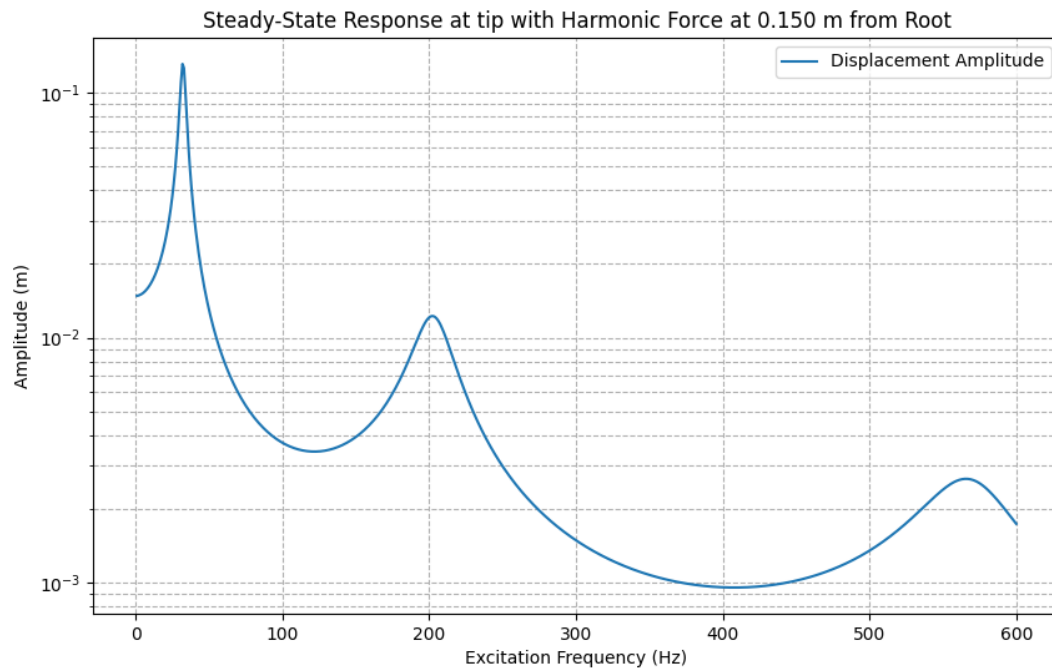


Figure 7: Steady-State Response at Tip due to Harmonic Force

The above figure depicts the displacement at the tip of the beam swept over the range of frequencies between 0-600 Hz for the harmonic loading of 1000N.

Maximum tip displacements of 13.3cm, 1.24cm and 0.27cm are observed at the first, second and third natural frequencies of the cantilever beam.

The overall maximum is at 31.4Hz and corresponds to 13.3cm of vertical displacement.

Question 4

```
# print(f"Addition of 1 kg mass at Motor")
# M Og[76,76] += 100
# K_ref, M_ref = apply_boundary_conditions(K Og, M Og, fixed_dofs)
# freqs_ref = solve_natural_frequencies(K_ref, M_ref, 4)
# response = frequency_response_analysis_with_motor(
#     K_red=K_ref,
#     M_red=M_ref,
#     C_red=C_ref,
#     node_idx=-2,
#     motor_position=motor_position,
#     motor_force_amplitude=motor_force_amplitude,
#     fixed_dofs=fixed_dofs,
#     total_nodes=n_ref
# )
```

Figure 8: Mass added to Cantilever Beam Code

Code is commented out since it did not work in simulation and other alternatives were pursued. A 1kg and 5kg mass were placed at motor location; this would lead to damping of the second and third natural frequencies but still leave the peak displacement of tip at the first natural frequency in excess of 8cm as shown in the figure below:

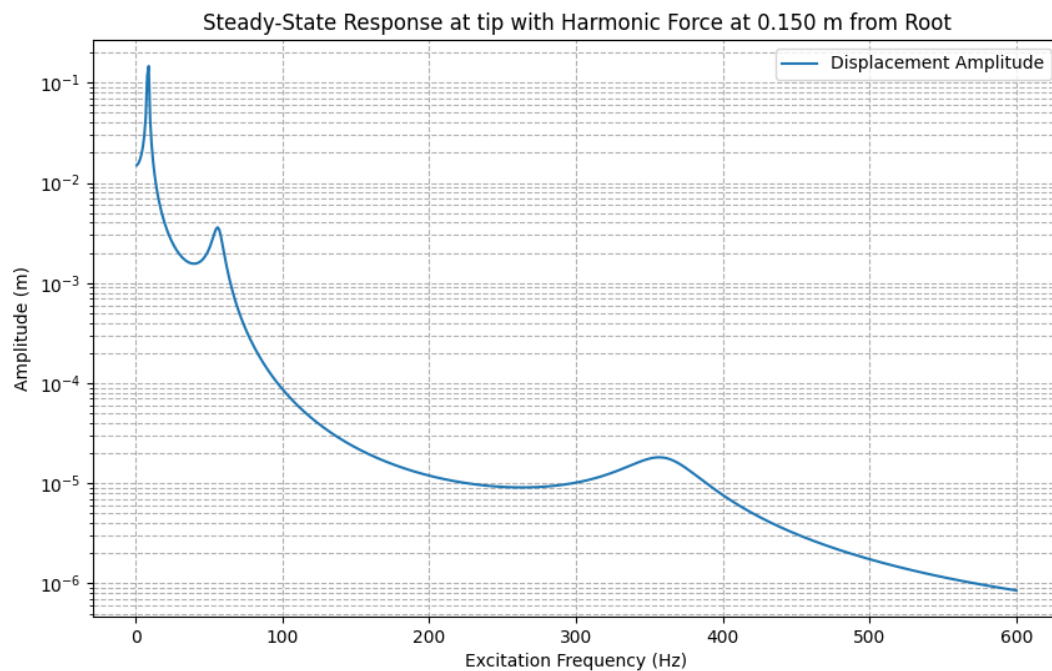


Figure 9: Steady State Response at Tip due to Harmonic Force with added Mass

Next, a cheap and accessible compression spring with suitable maximum load and spring constant was sourced from McMaster-Carr. The properties are tabulated in the figure below. [4]

Spring Type	Compression
System of Measurement	Inch
Length	5"
OD	2.906"
ID	2.344"
Wire	
Diameter	0.281"
Compressed Length @ Maximum Load	2.107"
Maximum Load	247 lbs.
Spring Rate	90 lbs./in.

Figure 10: Compression Spring Characteristics

According to the sizing guide, $90 \frac{lb}{in}$ corresponds to a metric spring constant of approximately $15750 \frac{N}{m}$. The maximum load of 247lbs corresponds to a metric force of 1100N. First placing the spring at the motor was tested:

```
print(f"Addition of 15750 N/m Spring at Motor")
spring_k = 15748
spring_n = 38
n_ref+=1
fixed_dofs = [0, 1, 256,257]
K_og, M_og = assemble_global_matrices(n_ref, elems_ref, E, I, rho, A, L_ref)
K_og = add_spring_global_k(K_og, M_og, spring_k, spring_n, elems_ref)
K_ref, M_ref = apply_boundary_conditions(K_og, M_og, fixed_dofs)
freqs_ref = solve_natural_frequencies(K_ref, M_ref, 4)
response = frequency_response_analysis_with_motor_modified(
    K_red=K_ref,
    M_red=M_ref,
    C_red=C_ref,
    node_idx=-2,
    motor_position=motor_position,
    motor_force_amplitude=motor_force_amplitude,
    fixed_dofs=fixed_dofs,
    total_nodes=n_ref
)
```

Figure 11: Addition of Spring to Cantilever Beam Code

The following curve was obtained as a result:

Addition of 15750 N/m Spring at Motor
Maximum displacement: $4.00144e-02+0.00000e+00j$ meters

Figure 12: Maximum Displacement at Tip with Spring Added

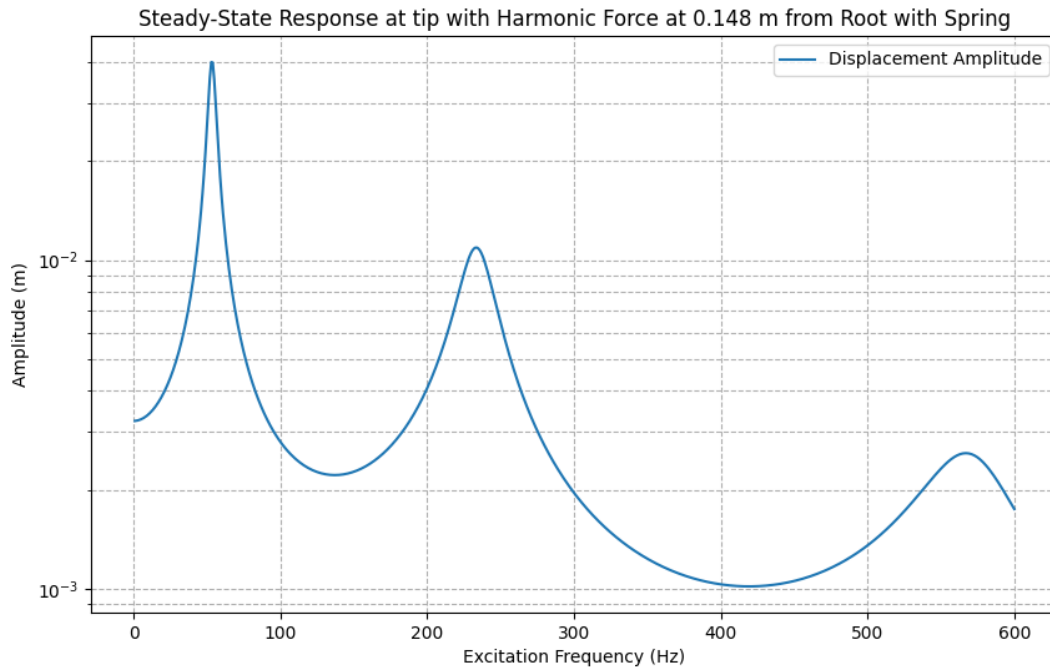


Figure 13: Steady State Response at Tip due to Harmonic Force with added Spring

A maximum tip displacement of 4cm was obtained in the vicinity of the first natural frequency which is well within our design constraint of 8cm or less displacement.

The spring only costs \$19.22 / unit and is a very cost-effective solution.

Conclusion

Just like any other finite element application; in the harmonic analysis of a cantilever beam the more refined a mesh is i.e. the greater the number of elements the closer the simulated results are to exact solutions.

Using a computer to test the cantilever beam forced harmonic response over 600 different frequencies is much faster than doing the same on paper and is really why finite element analysis shines as an engineering tool.

Simulation results are to be taken with a grain of salt, bad input lead to bad outputs so it is important to verify ones simulation results with hand calculations / relying on past experience / intuition / asking another engineers input.

References

- [1] Daryl L. Logan. 2000. A First Course in the Finite Element Method Using Algor (2nd. ed.). Brooks/Cole Publishing Co., USA.
- [2] OpenAI. (2023). *ChatGPT* (June 16 version) [Large language model]. <https://chat.openai.com/chat>
- [3] Engineeringtoolbox, "Cantilever beam calculations: Formulas, loads & deflections," Engineering ToolBox, https://www.engineeringtoolbox.com/cantilever-beams-d_1848.html (accessed Apr. 6, 2025).
- [4] Carr, McMaster, <https://www.mcmaster.com/96485K225/> (accessed Apr. 6, 2025).

Appendix A

Running Python Code

Use at least python 3.11 with scipy, matplotlib and numpy pre-installed and run the included .py file in your IDE of choice.